No Refund for the Token Tax: Speculative Decoding Across 21 Languages

Hanyu Yang (ORCID 0009-0005-0419-4070)

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Abstract

Tokenizers impose a token tax on non-English text: a Khmer sentence uses 4.7× more Qwen tokens than its English counterpart. Speculative decoding might pay part of it back. Because tokenizers split words into multiple segments, later segments should be easier to predict, allowing a lightweight drafter to achieve higher acceptance in high-fertility languages. We tested this on 21 languages using the built-in drafters of Qwen3.5-9B and Gemma 4 under fixed decision rules; there was no refund — English got the most tokens accepted per step. While continuation tokens offer a slight boost, the drafters guess worse outside English for almost every kind of token. In real serving with vLLM, speculation makes Gemma's English replies 2.4× faster than baseline, whereas Finnish replies are only 1.2× faster, expanding Finnish latency from 1.6× to 3.0× the English latency. The data and code are publicly available.

1 Introduction

The token tax. Language models process tokens rather than words. Tokenizers stem largely from English text, so other languages are cut into different, smaller pieces. On FLORES-200, a Khmer sentence takes 4.7 times as many Qwen3.5 tokens as its English source, and an Amharic one 4.5 times. Petrov et al. [1] and Ahia et al. [2] demonstrated consequences such as higher per-token fees, slower responses, and reduced usable context.

The hope. According to Google, inference made up about 60 percent of ML compute energy in 2019–2021 [3]. Workloads generate long outputs (reasoning traces, agents, and RL rollouts), among which rollouts dominate post-training time [4, 5]. Speculative decoding is the standard solution: a smaller model drafts several tokens, and the larger target model evaluates them in a single forward pass [6, 7]. Serving frameworks such as vLLM and SGLang support the feature, and DeepSeek-V3, Qwen3.5 and Gemma 4 provide built-in drafters [8, 9, 10].

The fertility refund. High-fertility languages—those requiring many tokens per word—might recover part of their tax. Because tokenizers split words into multiple segments, later segments are easier to predict once the first piece is known. The drafter should handle those continuations reliably, enabling the target model to accept more tokens per verification step. We call this hypothesis the *fertility refund*.

What we find. There is no refund; there is a surcharge. Evaluating two built-in drafters across 21 languages through exact emulation and pre-written decision rules, acceptance length (AL) did not increase with fertility; it stayed flat for Qwen and fell for Gemma. All 20 non-English languages ranked lower than English (Figure 1).

- The mechanism: Non-English text has many more continuation tokens, and these are easier to draft. That mechanism is real, but its impact is modest. Drafters lose far more because, outside English, they guess worse on almost every kind of token: word starts, continuations, and the rest (Figure 2).
- Real serving: The gap translates directly into wall-clock time. In vLLM serving, English benefits the most from speculation: Gemma achieves a speedup of 2.38× for English and 1.23× for Finnish. Without speculation, generating a Finnish response takes 1.6× as long as English; with speculation, Finnish takes 3.0× as long (Table 1, Figure 3).

Related work. Earlier evaluations noted diminished speculative acceleration beyond English. Spec-Bench's German→English translation showed the smallest speedup gains [11]. Yi et al. [12] and Do et al. [13, 14] designed drafters tailored to specific languages and measured individual execution times. Sandler et al. [15] and Paudel et al. [16] measured per-language acceptance using separate draft models; Paudel et al. used a Qwen3.5-9B verifier to evaluate 11 low-resource languages. None of these prior works examine fertility as a variable or evaluate its return on investment across languages. We investigate this directly for built-in drafters against English, using emulation and timed serving.

Why it matters. Speculative decoding benchmarks focus heavily on English benchmarks [11], meaning reported speedups overstate benefits for users working in other languages. The correction is inexpensive because teams can train drafters on multilingual target outputs and report speedups for each language.

Contributions.

1. Exact measurements of speculative acceptance across 21 languages for two built-in drafters, guided by decision rules set in place before the runs.
2. An exploratory decomposition showing that while the token-mix refund exists, the drafting deficit within the same token type outweighs it.
3. Real-world validation: we ran vLLM latency and sampling tests and evaluated two EAGLE-3 heads, confirming that the surcharge persists in serving.
4. Open artifacts: all raw records, analysis code, and a self-checking reproduction script are available at doi:10.5281/zenodo.23331652.

2 Background

Speculative decoding. A drafter proposes K candidate tokens. The target evaluates all K tokens in a single forward pass, retains the longest prefix it agrees with, and then appends one token of its own. Under greedy decoding, the output is unchanged [6, 7]. Acceptance length (AL) is the average number of tokens one verification step produces. Because a step that confirms a draft tokens produces $a + 1$ tokens, the acceptance length satisfies $\mathrm{AL} \geq 1$.

Built-in drafters. DeepSeek-V3 and Qwen3.5 train their multi-token prediction (MTP) head jointly with the model [8, 9]. Gemma 4 includes a separate MTP drafter that functions as an "assistant" checkpoint [10]. EAGLE-3 heads are trained after extracting the target features [17].

Fertility. We quantify fertility as the token ratio T_L/T_{en}: the token count for a sentence in language L divided by the token count for its English source, evaluated with the target's own tokenizer.

3 Setup

- Models: We evaluate Qwen3.5-9B, which features a single-layer MTP head applied recursively, alongside Gemma 4 E4B-it with its dedicated MTP drafter. Both run in bf16 on a single A100 at the pinned revisions.
- Data: The FLORES-200 devtest consists of 200 sentences in each of 21 languages, selected to represent fertility; the task is translation from English into each language, with the English targets originating from French translations.
- Measurement: The target first generates its own greedy translation y. We then run the drafter at every position of y to depth 5 and count how many drafts the target accepts for $K = 1, \ldots, 5$, producing exact AL values free of timing noise.
- Check against vLLM: The English AL of vLLM at $K = 3$ is within 4% of the emulated value for both models. Without speculation, vLLM greedy outputs match Hugging Face outputs on all 20 test prompts per model.
- Decision rules, set in place before the runs:
  - F1: the partial Spearman rho between AL at $K = 3$ and the ratio T_L/T_{en}, controlling for translation quality, does not exceed 0.10.
  - F2′: the median gap between step-1 acceptance of continuation tokens and of word-start tokens remains below 0.20.
  - Every wiring component and validation gate needs to succeed.
  - The refund hypothesis is rejected if either rule fires across both model families.
  - We documented the rules before the runs but they lack external timestamps; one change came after data: run 1 failed Qwen's wiring check, and we swapped in a weaker comparison, which made that check easier to pass.
- Serving: We deployed vLLM 0.31.0 on a single A100 40GB:
  - latency at batch size 1 across six languages, 30 sentences each;
  - sampling at $T \in \{0, 0.7, 1.0\}$ across all 21 languages, 100 sentences each;
  - two EAGLE-3 heads on Qwen3-8B: a general head trained on English chat data, and a "multilingual" head trained only on Hindi, Gujarati, and English.

4 Results

4.1 No refund

All gates pass in both families, and both rules fire in both.

- Qwen:
  - F1 $\rho = -0.033$: flat.
  - F2′ $= 0.172$: continuation tokens help, but by less than the 0.20 bar.
- Gemma:
  - F1 $\rho = -0.656$: acceptance falls with fertility.
  - F2′ $= 0.049$.
- AL at K = 3:
  - Qwen: English 3.16, other languages 1.89–2.81.
  - Gemma: English 2.93, other languages 1.26–2.19.
- Refund fraction: It is negative wherever it is defined: Qwen -0.86 to -0.10, Gemma -1.65 to -0.66.

Qwen's assessment is based on point estimates; its language-bootstrap intervals cross both thresholds (F1 $[-0.53, 0.52]$, F2′ $[0.152, 0.224]$).

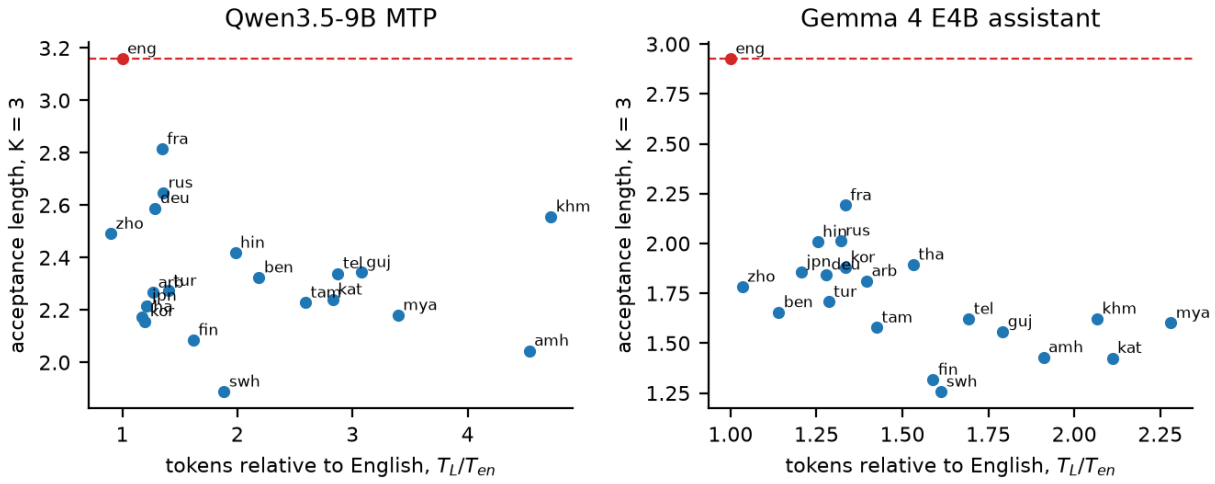


Figure 1: AL at $K = 3$ against the token ratio, one point per language. The dashed line is English.

4.2 Why: the mix helps, but the drafter reduces acceptance

We partition each language's step-1 acceptance gap relative to English into two components using a two-fold Oaxaca decomposition:

- Mix: The portion attributable to different token-type distributions, measured by English acceptance rates.
- Within-type: The portion attributable to lower drafter acceptance within the same token type.

The results:

- The mix component is small and positive: on average over 20 languages, it adds 2.6 percentage points for Qwen and 5.5 for Gemma.
  - Non-English text contains substantially more continuation tokens, ranging 0.19–0.82 versus English 0.08 for Qwen, and 0.16–0.87 versus 0.06 for Gemma.
  - So the refund mechanism is real, but small.
- The within-type drafting deficit is large and negative: -19.5 percentage points (Qwen) and -42.6 (Gemma), remaining below zero across all 20 languages.
- The deficit persists with depth: When draft 1 is accepted, Qwen's draft 2 is accepted 46–74% of the time outside English, against 85% in English (for Gemma, 43–66% against 83%).

4.3 In milliseconds

Table 1. Speedup at $K = 3$, batch size 1, vLLM 0.31.0, one A100 40GB.

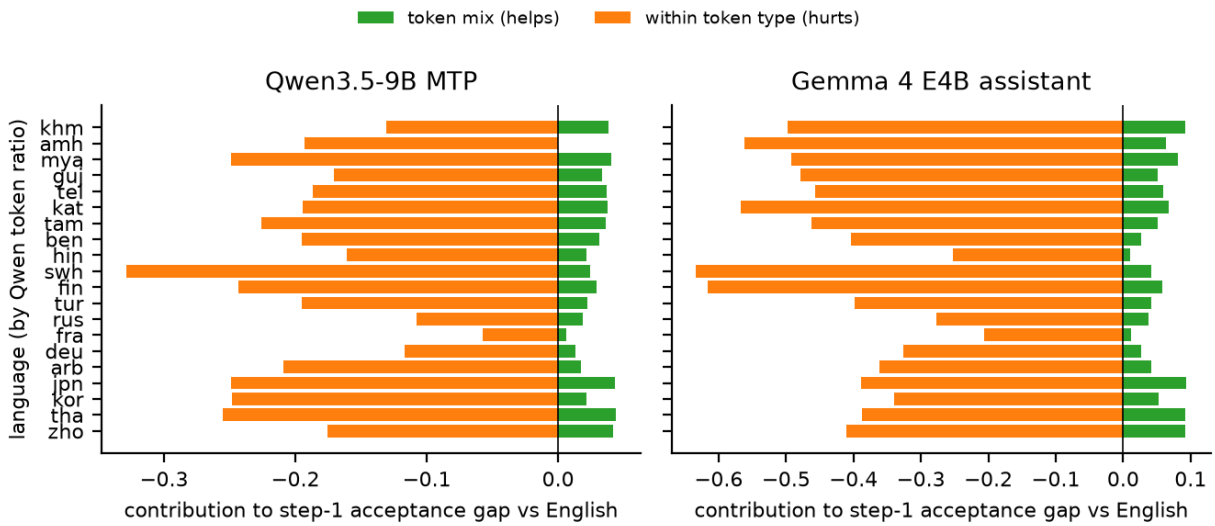


Figure 2: Each language's step-1 acceptance gap to English, split into the token-mix part and the within-type part. Languages are ordered by Qwen's token ratio.

speedup at K = 3	eng	zho	rus	fin	hin	amh
Qwen3.5-9B MTP	1.94	1.60	1.73	1.52	1.65	1.41
Gemma 4 E4B MTP drafter	2.38	1.54	1.86	1.23	1.78	1.30
Qwen3-8B + EAGLE-3 (general)	1.97	1.03	0.99	1.01	1.08	0.92*
Qwen3-8B + EAGLE-3 (multilingual)	1.31	0.92	0.95	0.93	1.64	1.25*

3 valid items.

- The built-in drafters: For both, every non-English language gets a smaller speedup than English, with non-overlapping intervals.
- The gap widens: The latency premium over English grows under speculation. Gemma's Finnish goes from 1.55× English latency to 3.02×. Across the six languages, speedup falls with fertility ($\rho = -0.54$ for Qwen, -0.71 for Gemma; n = 6).
- The general EAGLE-3 head gives almost nothing outside English (0.92–1.08).

4.4 Sampling and trained drafters

Table 2. Spearman ρ(AL, T_L/T_{en}) over 21 languages, $K = 3$.

	T = 0	T = 0.7	T = 1.0
Qwen3.5-9B MTP	−0.21	−0.38	−0.48
Gemma 4 E4B MTP drafter	−0.68	−0.67	−0.67
EAGLE-3 general	−0.61	−0.65	−0.72
EAGLE-3 multilingual	0.02	−0.02	−0.02

- Sampling makes Qwen's trend more negative: English keeps the highest AL at every temperature for both built-in drafters.
- The general EAGLE-3 head falls with fertility, like the built-in drafters.
- The multilingual EAGLE-3 head is the exception:
  - Over all 21 languages its ρ is about 0.

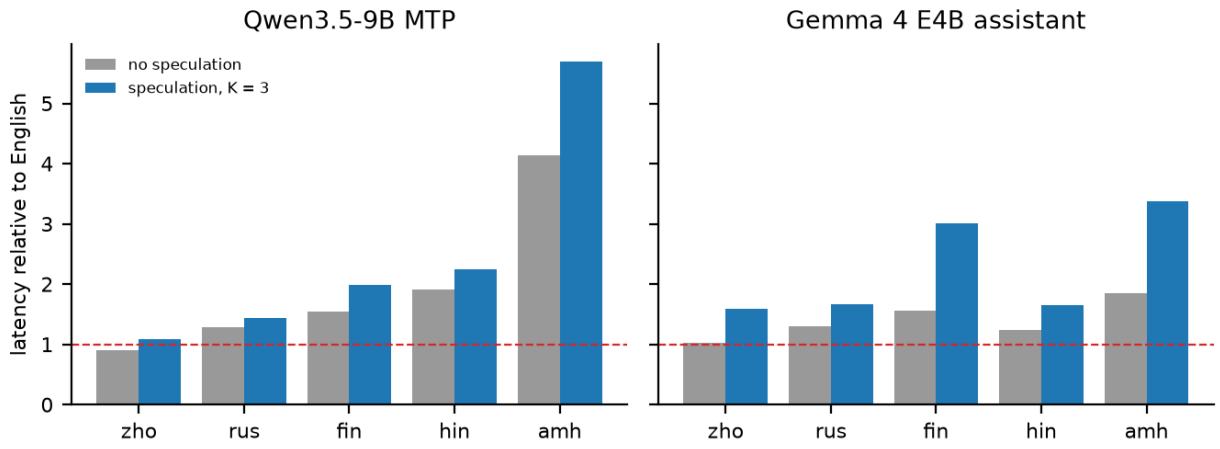


Figure 3: Latency relative to English, without speculation (grey) and with it at $K = 3$ (blue). The dashed line is parity with English.

  - Qwen3-8B produces degenerate (capped or looping) output for many sampled items in a few languages: 83 of 100 Amharic items at T = 0, and about half of Burmese. vLLM's counters cannot exclude those items.
  - Dropping languages above an invalid-item threshold between 5% and 50%, its ρ ranges from +0.08 to +0.42 at T = 0 and from −0.13 to +0.15 at T = 1.0. The sign depends on the threshold. The general head stays at −0.57 to −0.73 at every threshold.
  - Its English AL is 1.70, against 2.42 for the general head. It beats the general head in only 4 of 21 languages, mostly Gujarati (+1.20) and Hindi (+0.65).
- Takeaway: The one head trained on non-English data shows no downward trend, from a weak English start. This is post hoc, one head on one target, and its training prompts came from FLORES-200.

5 What this means

- Report speedups per language: A single English number overstates the gain for most users. Our six-language latency panel took about one GPU-hour.
- Drafters are a fairness lever: The within-type gap puts the bottleneck in the drafter more than in the tokenizer. Training a drafter on multilingual target outputs is cheap next to retraining a tokenizer, and Yi et al. [12] and Do et al. [13] show it works.
- Inference-heavy workloads may amplify the gap: Speculative decoding is already used to speed up RL rollouts [4, 5]. If the per-language gap carries over to long reasoning traces, a speed gap becomes a cost gap. We measured translation only.

6 Limitations

- Scope:
  - One task (translation), FLORES sentences, two built-in drafters plus two EAGLE-3 heads on a third target.
  - One engine version and one GPU type. Latency is at batch 1 in six languages.

- Statistics:
  - Sampling AL is a per-language aggregate that includes invalid items. This is negligible for Qwen and Gemma (35 and 3 of 6,300) but not for the Qwen3-8B EAGLE arms (419 and 420 of 6,300).
  - The decomposition and the threshold sweep were chosen after the verdict.
- The multilingual EAGLE-3 head: Its training data, per its model card, are Qwen3-8B responses to FLORES-200 prompts in Hindi, Gujarati and English. The card does not name the split. Its Hindi and Gujarati gains may partly reflect overlap with our test set.
- The decision rules: They were stated before the runs but are not externally timestamped, and one gate amendment followed the first run.
- The vLLM serving numbers: They come from one run. In vLLM, speculative greedy output is often not identical to plain greedy output (down to half of Qwen's Amharic outputs at $K = 1$), so speedups are latency ratios over the same inputs.

Data and code

Everything is at doi:10.5281/zenodo.23331652 (CC BY-SA 4.0; code MIT), mirrored on GitHub. `reproduce.sh` recomputes the verdict, the decomposition and the serving analyses from the raw records, and fails loudly on any mismatch.

Disclosure

This paper is AI-assisted. Claude Opus 5.5 (Anthropic) wrote the measurement, serving and analysis code and ran the experiments under my direction, and I checked the results. The data deposit was reviewed by GPT-6 Luna (OpenAI, through Codex), Gemini 3.1 Pro (Google) and a second Claude Opus 5.5 instance. Claude Opus 5.5 checked the citations, made the figures, and drafted the prose from my notes. I rewrote and edited that draft. I take responsibility for the paper.

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